

Julian Pietraroia

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EDUCATION

University of British Columbia — GPA: 3.9 **Expected April 2027**

Bachelor of Applied Science in Manufacturing Engineering

- Dean's Honour List — Top 10% of Faculty of Applied Science
- Relevant Coursework: Industrial Automation, Manufacturing Processes, Mechanics of Materials, Mechanical Vibrations, Engineering Measurements & Instrumentation, Machine Dynamics, Software Design for Engineers

EXPERIENCE

Hardware Engineer — Stealth Startup (\$25M raised), San Francisco, CA **June 2026 – Present**

- Designed and manufactured a 1-DOF UMI (Universal Manipulation Interface) gripper now used by 200+ data collectors to scalably capture robot manipulation demonstrations
- Designed a full 22-DOF articulating glove that individually tracks every finger joint for high-fidelity hand-pose and teleoperation data
- Developed and deployed a custom 1- and 2-DOF actuated gripper across teleoperated data collection stations to gather 1,000 hours of manipulation data

Manufacturing Engineering Intern — cellcentric (Toyota, Volvo & Daimler Truck), Burnaby, BC **May 2024 – Aug 2025**

- Applied Lean Manufacturing to optimize a state-of-the-art fuel cell production process, reducing cost and improving cycle times by 40%
- Led Value Stream Mapping to identify bottlenecks and calculate process capacities, driving continuous-improvement initiatives
- Performed data analysis and root cause analysis on production data to surface trends and predict failure points
- Built company-wide Power BI dashboards for production and budget reporting, improving efficiency and reducing reporting time

Mechanical Lead — Ten Ton Robotics (VEX U Competitive Robotics), West Vancouver **Aug 2023 – Present**

- Led mechanical design of a two-robot VEX U system that won the 2025 World Championship and back-to-back Skills World Championships against 300+ university teams
- As Mechanical Lead, guided a 14-member team through the full design cycle, from concept and CAD to fabrication and iteration
- Applied both subtractive (CNC) and additive (3D printing) manufacturing to produce unique, high-performance components
- Conducted FEA in Ansys on custom parts to optimize structural integrity and performance

PROJECTS

Line Following Robot **Apr 2026 – May 2026**

- Designed and manufactured a fully autonomous line-following robot with obstacle avoidance that generates a heat map of its environment
- Implemented sensor fusion of encoder and IMU odometry to track robot position and orientation during autonomous navigation
- Integrated IR-based line detection and proximity sensing for reliable autonomy across varied environments

One Pound Combat Robot — Personal Project **Apr 2023 – June 2025**

- Engineered a competitive combat robot in Fusion 360, selecting composite and metal materials (carbon-fiber nylon, TPU, titanium, AR-500 steel) for durability
- Used rapid prototyping and researched gyroscopic motion and center of gravity to engineer optimized, smooth movement

SKILLS

Mechanical:	Robotics, Mechatronics, GD&T, Design for Manufacturing (DFM)
Manufacturing:	Injection Molding, Thermoforming, CAM (Computer-Aided Manufacturing), SPC
Design / CAD:	SolidWorks, Onshape, Fusion 360, Ansys (FEA)
Electrical / Controls:	PLC, Microcontrollers, Firmware / Embedded Systems, Control Systems (PID), Soldering, Oscilloscope
Software:	Python, C / C# / C++, MATLAB, Git, Excel, Minitab, Ladder Logic, LaTeX

VOLUNTEER EXPERIENCE

Head Referee — Penguin Robotics, Richmond, BC **Sep 2022 – Present**

- Referee VEX Robotics competitions across British Columbia, helping create a competitive and fair environment for teams

AWARDS

VEX Robotics World Champions (2025) • VEX Robotics Excellence Award (2025) • VEX Robotics Skills Challenge World Champions (2024, 2025) • A.E. Coleman Leadership Award (2022) • Schulich Leader Nominee (2022)